

explored to reduce consumption and create a healthier indoor environment, says Amit Agrawal, Chief Business Officer of Web Werks-Iron Mountain Data Center, which has six centres in India and 20 globally. “We also prioritise the use of energy-efficient equipment throughout our data centres, including servers, cooling systems, and power distribution units, to minimise energy waste and lower operational costs,” he adds.

Additionally, data centres can pack more density into a given space by using higher-density hard disc drives (HDDs). Khalid Wani, Senior Director of Sales for the India region at the American storage drive maker, Western Digital, explains that if a data centre uses a 22 TB (terabyte) HDD instead of an 8 TB or 10 TB HDD, then by simple math it would need about 75% to over 93% less space to store the same amount of data.

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Of course, newer data centres—those that are being constructed now—can code sustainable solutions into the very design of the building. But there is a cost element attached. It varies based on factors like size, location, and technology, and incorporating green features could result in a 5–15% premium upfront, say experts.

“As India is fast emerging as a co-location market for data centres, and for constructing new data centres in India, there are typically two construction methods being used—horizontal construction with a pre-engineered building (PEB) structure and vertical construction with a concrete structure,” says Anshuman Magazine, Chairman & CEO, India, South-East Asia, Middle East & Africa, at real estate services firm CBRE. “The fit-out and MEP (Mechanical, Electrical, and Plumbing) scope are executed in phases based on the data centre’s specific requirements. Modular construction techniques are gaining traction for data centres,” he adds. The PEB approach is seen to be better

as it involves building data centre modules off-site and assembling them on-site, offering advantages such as faster deployment, scalability, and cost-effectiveness.

But co-locating data centres face quite a few challenges.

“These challenges include coordinating sustainability efforts across varied tenants, managing complex ownership structures, and addressing common operational problems,” says Amit Luthra, Managing Director—India, data centre solutions firm Lenovo Infrastructure Solutions Group.

But there are solutions out there in the market. Hewlett Packard Enterprise (HPE), as a part of its onboarding process, evaluates workloads and optimises the hardware and infrastructure accordingly to increase server utilisation, optimise server refresh cycles, and rightsize redundancy requirements. “Moreover, the metering capabilities delivered as part of HPE GreenLake services enable businesses to match their workloads and capacity closely to consumption, eliminating wasteful over-provisioning without incurring the risks of under-provisioning,” says Rajesh Dhar, Senior Director of the Industry Verticals at HPE India. The HPE GreenLake platform is a dashboard that offers insights into IT energy consumption, carbon emissions, and electricity costs.

“Over the long term, adopting green solutions may also help the data centre players optimise their operational costs and improve the RoI through solutions such as using green energy during peak hours, when non-renewable sources are expensive,” says Siddhartha Tipnis, Partner & Technology Sector leader at consultancy Deloitte in India.

Additionally, a lower carbon footprint enhances the brand image. Plus, there may be potential tax benefits or incentives available from government initiatives.

Thus, greening data centres can result in a win-win situation for businesses. **BT**

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**Web Werks-Iron Mountain Data Center prioritises energy-efficient equipment throughout its data centres, including servers, cooling systems, and power distribution units**